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Alexandra P. P. Natalense; Robert R. Lucchese



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Cross section and asymmetry parameter calculation for sulfur 1s photoionization of SF₆

Alexandra P. P. Natalense and Robert R. Lucchese

Department of Chemistry, Texas A&M University, College Station, Texas 77843-3255

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We use the polyatomic Schwinger variational method with Padé corrections to calculate the cross section and asymmetry parameter for the sulfur 1s core level photoionization of SF₆, for photon energies from threshold up to 2600 eV. Our results show very good agreement with experimental cross sections. Our asymmetry parameter show good qualitative agreement with experiment. A resonant feature at a photon energy of ≈ 2550 eV is found to be due to a nonvalence type resonant state which is trapped by an $l=9$ angular momentum barrier. © 1999 American Institute of Physics. [S0021-9606(99)00836-3]

I. INTRODUCTION

The complexity in the description of the final state and of the molecular many electron problem have restricted most of the theoretical work on photoionization of molecules to linear and small polyatomic targets. For linear molecules, a variety of methods and approximations have been employed. Recent results include the many-body calculation for CO using a second order perturbation expansion of the Coulomb interaction,¹ the multiconfiguration linear response theory applied to the photoionization of N₂, O₂, CO, and NO,² the algebraic diagrammatic construction formulation of the random phase approximation for H₂ and N₂,³ and several applications of the iterative Schwinger variational method using different approximations; frozen-core Hartree–Fock for N₂,⁴ relaxed-core Hartree–Fock for CO,⁵ the multichannel configuration-interaction approximation (MCCI) for CO,⁶ and C₂H₂,⁷ and a more general implementation of the MCCI approximation using a complete active space configuration-interaction for the N₂ molecule.⁸ The available theoretical results for polyatomic molecules are much more restricted. Some recent studies include the results of local density approximation for valence photoionization cross sections using the Stieltjes imaging technique for many linear molecules and SF₆, TiCl₄, and Cr(CO)₆,⁹ and a recent application of the one-center expansion to the photoionization of C₆H₆,¹⁰ the separated channel, frozen core calculations using the complex Kohn variational method for NH₃,¹¹ calculations of differential photoionization cross sections using the random phase approximation for PH₃,¹² and the static-exchange frozen core application of the iterative Schwinger variational method for valence photoionization of CH₄,¹³ H₂O,¹⁴ C₂H₄,¹⁵ and SiH₄.¹⁶

In this paper we show the first application of the polyatomic Schwinger variational method with Padé corrections¹⁷ to a core level photoionization problem for a polyatomic molecule with many heavy centers. This method takes advantage of the molecular symmetry and allows us to study highly symmetric molecules such as SF₆ with relative low computational effort. Our results for the S 1s core level photoionization of SF₆ are in good agreement with available

experimental data. We are also applying the local adiabatic static model exchange method (ASME) (Ref. 18) to resonances of electron–ion scattering for the first time.

II. METHOD

Both the initial neutral molecule electronic wave function and the final ionized molecule electronic wave function are represented by single determinants constructed using the Hartree–Fock orbitals of the initial neutral state. The final N -electron state (continuum photoelectron + molecular ion) can then be variationally obtained from

$$\langle \delta\Psi_{\mathbf{k}} | H - E | \Psi_{\mathbf{k}} \rangle = 0, \quad (1)$$

where $\delta\Psi_{\mathbf{k}}$ represent variations in the final state due to variations on the continuum wave function $\psi_{\mathbf{k}}(\mathbf{r})$, and $\mathbf{k}$ is the momentum of the ejected electron. The final photoionization problem is then reduced to solving the problem of an electron under the action of the potential of the ion. Thus, we do not consider many-electron effects. The following scattering equation can be obtained from Eq. (1) (Ref. 19) (in atomic units),

$$\left[-\frac{1}{2}\nabla^2 + V(\mathbf{r}) - \frac{k^2}{2} \right] \psi_{\mathbf{k}}^{(\pm)}(\mathbf{r}) = 0, \quad (2)$$

where $V(\mathbf{r})$ is the static exchange potential given by:

$$V(\mathbf{r}) = - \sum_{\gamma=1}^M Z_{\gamma} |\mathbf{r} - \mathbf{R}_{\gamma}|^{-1} + \sum_{i=1}^{n_{\text{occ}}} (2\hat{J}_i - \hat{K}_i) + \hat{J}_{\perp 5} + \hat{K}_{\perp 5} \quad (3)$$

for M nuclei of charge Z_{γ} located at $\mathbf{R}_{\gamma}$, where n_{occ} is the number of doubly occupied orbitals. $\hat{J}_i$ is the Coulomb operator,

$$\hat{J}_i(\mathbf{r}_1) = \int \frac{\phi_i^*(\mathbf{r}_2)\phi_i(\mathbf{r}_2)}{r_{12}} d^3r_2, \quad (4)$$

and $\hat{K}_i$ is the nonlocal exchange potential operator

$$(\hat{K}_i\psi)(\mathbf{r}_1) = \phi_i(\mathbf{r}_1) \int \frac{\phi_i^*(\mathbf{r}_2)\psi(\mathbf{r}_2)}{r_{12}} d^3r_2. \quad (5)$$

Correlation and polarization effects can be included in the calculations through the addition of a local, energy independent model correlation polarization potential, such as described in Ref. 10, but we found that they do not affect our final results for the S 1s photoionization problem in SF₆. Even though we are studying a core level photoionization problem, our approximation does not include the effect of relaxation of the ion due to the localized nature of the core hole, which is known to affect the position, width, and magnitude of resonances.²⁰

To solve the scattering problem, we use the single center expansion (SCE) method, where all three dimensional functions are expanded on a set of angular functions $X_{lh}^{p\mu}(\theta, \phi)$, which are symmetry adapted, according to the irreducible representations of the molecular point group (O_h for SF₆). An arbitrary three dimensional function $F^{p\mu}(r, \theta, \phi)$ is then expanded as

$$F^{p\mu}(r, \theta, \phi) = \sum_{lh} r^{-1} f_{lh}^{p\mu}(r) X_{lh}^{p\mu}(\theta, \phi), \quad (6)$$

where

$$X_{lh}^{p\mu}(\theta, \phi) = \sum_m b_{lhm}^{p\mu} Y_{lm}(\theta, \phi), \quad (7)$$

and p is one of the irreducible representations of the molecular point group, μ is a component of the representation p , and h indexes all possible $X_{lh}^{p\mu}$ belonging to the same irreducible representation ($p\mu$) with the same value of l . The radial functions $f_{lh}^{p\mu}(r)$ are represented on a numerical grid. When solving the scattering equations we enforce orthogonality between the t_{1u} continuum solutions and the occupied orbitals.¹⁹

The matrix elements of the dipole operator are

$$I_{\mathbf{k},\hat{n}}^L = (k)^{1/2} \langle \Psi_i | \mathbf{r} \cdot \hat{n} | \Psi_{f,\mathbf{k}}^{(-)} \rangle \quad (8)$$

for the dipole length form, and

$$I_{\mathbf{k},\hat{n}}^V = \frac{(k)^{1/2}}{E} \langle \Psi_i | \nabla \cdot \hat{n} | \Psi_{f,\mathbf{k}}^{(-)} \rangle \quad (9)$$

for the dipole velocity form, where $|\Psi_i\rangle$ is the initial bound state, $|\Psi_{f,\mathbf{k}}^{(-)}\rangle$ is the final continuum state, E is the photon energy, $\mathbf{k}$ is the momentum of the photoelectron, and $\hat{n}$ is the direction of polarization of the light, which is assumed to be linearly polarized.

The matrix elements $I_{\mathbf{k},\hat{n}}^{L,V}$ of Eqs. (8) and (9) can be expanded in terms of the $X_{lh}^{p\mu}$ functions of Eq. (7) as¹⁴

$$I_{\mathbf{k},\hat{n}}^{(L,V)} = \left[\frac{4\pi}{3} \right]^{1/2} \sum_{p\mu lh\nu} I_{lh\nu}^{p\mu(L,V)} X_{lh}^{p\mu}(\hat{k}) X_{1\nu}^{p\nu\mu\nu}(\hat{n}). \quad (10)$$

The differential cross section is given by

$$\frac{d\sigma^{L,V}}{d\Omega_{\mathbf{k}}} = \frac{\sigma^{L,V}}{4\pi} [1 + \beta_{\mathbf{k}}^{L,V} P_2(\cos \theta)], \quad (11)$$

where the asymmetry parameter can be written as¹⁴

$$\begin{aligned} \beta_{\mathbf{k}}^{L,V} = & \frac{3}{5} \frac{1}{\sum_{p\mu lh\nu} |I_{lh\nu}^{p\mu(L,V)}|^2} \\ & \times \sum_{p\mu lh\nu m m_\nu} (-1)^{m' - m_\nu} I_{lh\nu}^{p\mu(L,V)} \\ & \times (I_{l'h'\nu'}^{p'\mu'(L,V)})^* b_{lhm}^{p\mu} b_{l'h'm'}^{p'\mu'} b_{1\nu m_\nu}^{p\nu\mu\nu} b_{1\nu'm_\nu'}^{p'\nu\mu'\nu'} \\ & \times [(2l+1)(2l'+1)]^{1/2} (1100|20)(l'l00|20) \\ & \times (11-m'_\nu m_\nu | 2M')(l'l-m'm | 2-M'), \end{aligned} \quad (12)$$

and the $(l'l m'm | L'M')$ are the usual Clebsch–Gordan coefficients. The total cross section is

$$\sigma^{L,V} = \frac{4\pi^2}{3c} E \sum_{p\mu lh\nu} |I_{lh\nu}^{p\mu(L,V)}|^2, \quad (13)$$

where c is the speed of light.

III. COMPUTATIONAL DETAILS

Our Hartree–Fock molecular target was calculated using the 6-311G basis set from the GAUSSIAN 94 code, augmented with two d polarization functions. The self-consistent field energy obtained with this basis set was $-994.219\,509$ hartree. Our calculations were carried out in the fixed-nuclei approximation with a S–F distance of 1.561 Å.²¹ In the O_h point group all three components of the dipole operator transform as the T_{1u} irreducible representation, so that we only needed to compute the final continuum states $|\Psi_{f,\mathbf{k}}^{(-)}\rangle$ belonging to the T_{1u} symmetry to calculate the matrix elements of Eqs. (8) and (9). The experimental ionization potential for the sulfur 1s ionization is 2490 eV.²²

Our partial wave expansion for the molecular orbitals and for the scattering wave functions included up to $l=30$. With this truncation, the largest error in the normalization of the molecular orbitals was 6% for the $1t_{1u}$ orbital (the 1s on the F atoms), and the smallest error was for the $1a_{1g}$ one, which was correct up to the 6th decimal place.

IV. RESULTS AND DISCUSSION

Next we show our results for the sulfur 1s photoionization of SF₆, compared to the available experimental data²³ and to the multiple scattering model results of Ref. 24. Our calculations using either the length form or the velocity form of the dipole matrix elements produced the same results for the cross section and asymmetry parameter. The photoionization cross section, shown in Fig. 1, presents two main features: the first one is a maximum that appears in the experimental cross section at a photon energy of 2517 eV and in our calculation at 2518 eV. The second feature is a resonance, as indicated by the behavior of the eigenphase sum of Fig. 2. The peak in our calculated cross section is at a photon energy of 2549 eV, and the position of the experimental peak is 2555 eV. To obtain the width (Γ) and the photon energy (E_R) of this broad resonance our eigenphase sum was fitted to a Breit–Wigner expression using the functional form²⁵

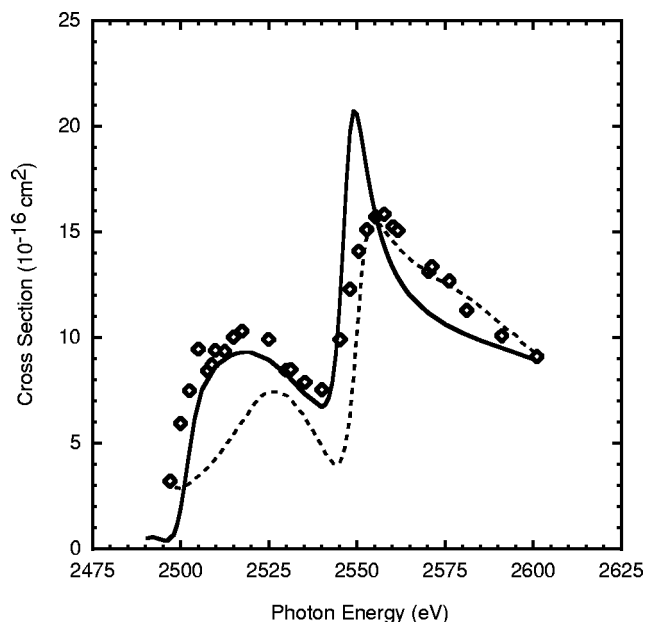


FIG. 1. Cross sections for S 1s photoionization of SF₆. Solid line, present result; dashed line, multiple scattering results (Ref. 24); diamonds, experiment (Ref. 23).

$$\delta(E) = a + b(E_R - E) + c(E_R - E)^2 + \tan^{-1} \left[\frac{\Gamma}{2(E_R - E)} \right]. \quad (14)$$

The fitting produces a resonance photon energy of $E_R = 2547.1$ eV, and a width of $\Gamma = 8.5$ eV. Our calculated cross section shows a resonance that is narrower than the experimental one, about 5×10^{-16} cm² more intense and the peak position appears at an energy 6 eV lower than the experimental peak. These discrepancies are probably due to the fact that our calculations do not include the relaxation of the ionic orbitals, which produces a screening of the 1s core

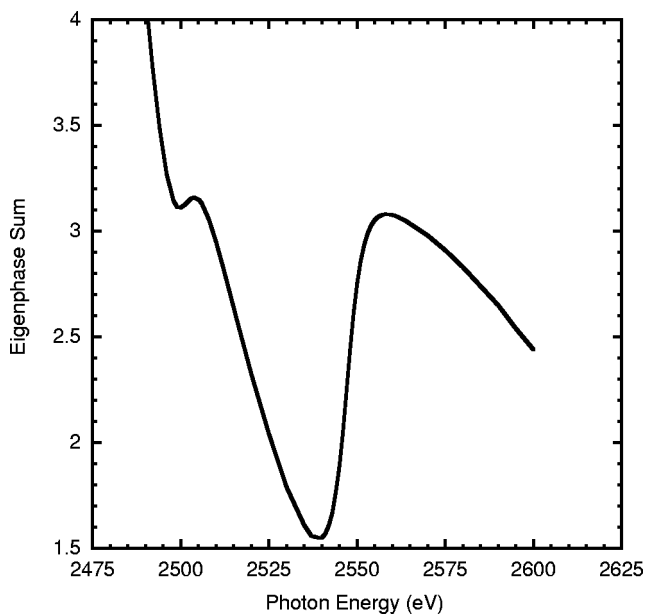


FIG. 2. Eigenphase sum for the T_{1u} scattering from SF₆⁺ in the static exchange approximation.

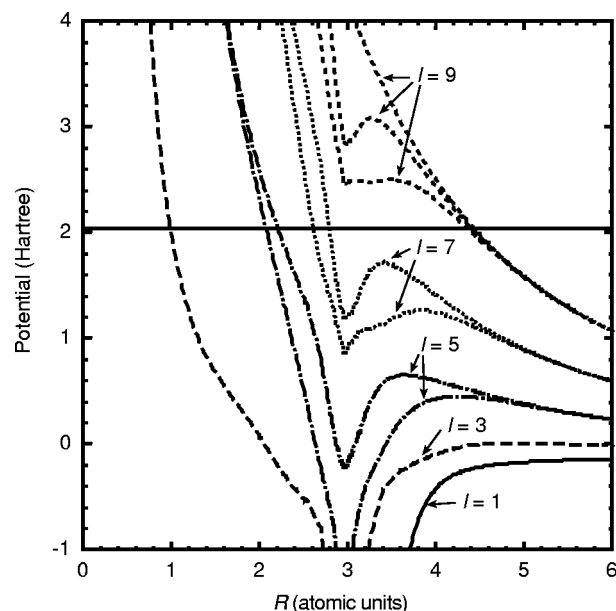


FIG. 3. Local adiabatic static model exchange potentials for partial waves up to $l=9$. Solid line, $l=1$; long dashed line, $l=3$; dotted-dashed line, $l=5$; dotted line $l=7$; short dashed line, $l=9$. The horizontal line approximately at 2.04 hartree indicates the ASME resonance energy E_R (55.5 eV photoelectron energy or 2545.5 eV photon energy).

hole. The lack of description of this effect produces a potential that is too attractive, resulting in the discrepancies mentioned above.²⁰ The multiple scattering result shows very good agreement with the experiment at the resonance region, but for photon energies below 2550 eV the agreement is only qualitative. This discrepancy may be due to the approximations for the molecular potential in the multiple scattering theory,²⁴ since the details of the actual potential are more important for lower electron energies.

In order to study the qualitative features of this resonance, we use the same method described in Ref. 18 to study shape resonances in electron-molecule scattering. The basic idea is to use a simplified model potential that allows a clearer understanding of the main features of the resonant process, and, within a good approximation, includes sufficient details of the full scattering problem. We have then used the purely local adiabatic static model exchange potential (ASME), which also does not include orthogonality constraints and was successfully applied in Ref. 18 to the analysis of the resonant states in electron scattering from SF₆ molecules.

In Fig. 3 we show the resulting adiabatic radial potentials for partial waves up to $l=9$. The horizontal line placed at a photoelectron kinetic energy of 2.04 hartree ($h\nu \approx 2545.5$ eV) shows the energy of the resonance calculated with the ASME potential. The corresponding width in this approximation was 7.5 eV. The results in this figure imply that the only channel which can provide a trapping mechanism is the $l=9$ channel. Figure 4 shows the squared wave function for the resonant state for each partial wave using the ASME potential. The squared wave function for $l=3$ was normalized to 1 at 1.5 a. u. This figure shows only the results that were higher than 0.05 at large values of R . The position

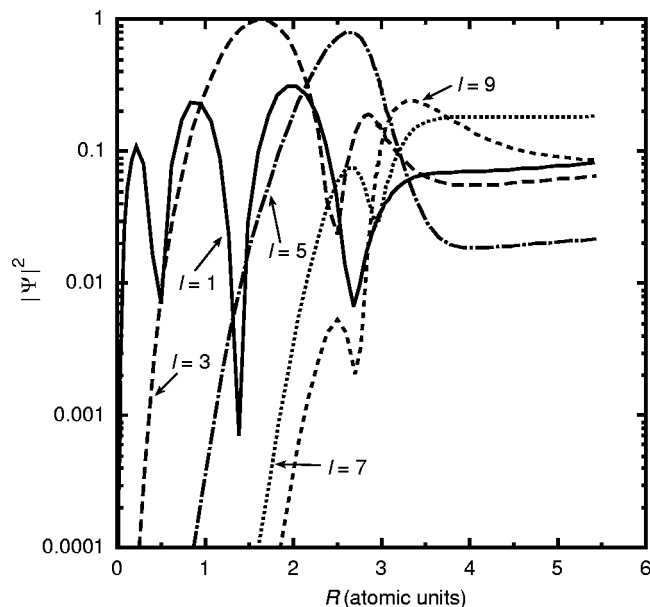


FIG. 4. Partial wave decomposition of the squared ASME wave function for the resonant state. The lines are labeled as in Fig. 3.

of the fluorine atoms is about 2.9 a.u. Figure 4 shows that several partial waves give important contributions to the wave function for smaller values of R , but, in the following discussion we will concentrate on the region beyond the atomic centers, which is important in determining the trapping mechanism. In the region right beyond the fluorine atoms, i. e., between 3 a.u. and 4 a.u., the most important partial wave contribution comes from $l=9$, which decays nearly linearly for larger values of R , indicating a tunneling through the potential barrier in the $l=9$ adiabatic potential shown in Fig. 3. It is interesting to note that in the asymptotic region the dominant partial wave is no longer $l=9$, but $l=7$. This suggests that the resonant state decays by scattering to the lower partial wave $l=7$ in which the potential barrier is lower than 2.04 hartree (see Fig. 3), rather than decaying purely through tunneling in the $l=9$ channel. Figure 4 thus indicates that there is a strong mixing of $l=7$ and $l=9$ partial waves in the resonant state in the region outside the nuclei.

Figures 5 and 6 show the contour plots of the real part of the resonant wave function. These plots show “cuts” of the wave function at the planes shown in Fig. 7. Figure 5 is related to plane a , which is the xy plane and contains the sulfur atom at the origin and four fluorine atoms. The radial lines extending out of the plot frame of this figure indicate the positions of the nodes of a purely $l=9$ function. The dotted lines show the approximate nodes of our wave function, which present a strong $l=9$ behavior in the region just outside the fluorine atoms. The parabolic nodes that are symmetric with respect to the x axis in Fig. 5 resemble previous results for N₂ (Ref. 18) and other linear molecules.²⁶ This idea is more clearly illustrated by Fig. 6, which shows a “cut” of the wave function according to plane b of Fig. 7. This plane includes only the sulfur atom and the two fluorine atoms along the x axis and makes 45° with the y direction. The radial lines in Fig. 6 indicate the positions of the nodes

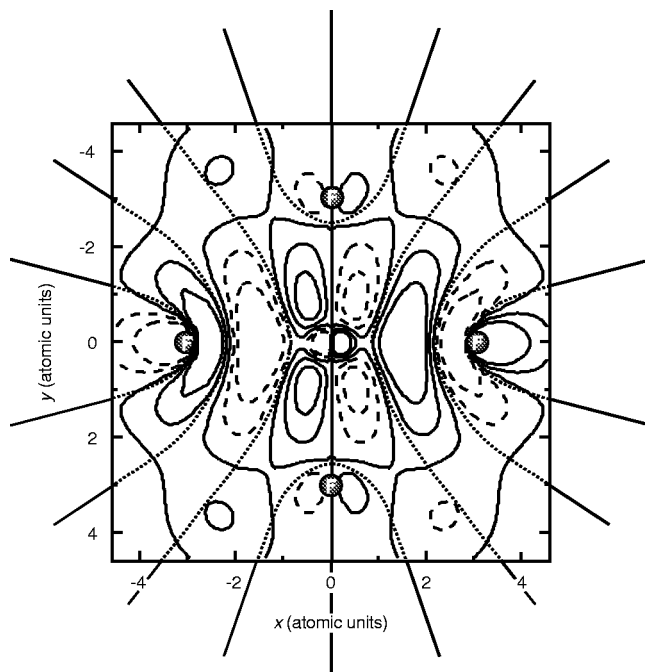


FIG. 5. Contour plot of the ASME resonant state in the plane a of Fig. 7 (plane xy). Solid line, positive contours; dashed line, negative contours; dotted lines, approximate location of the nodes. The radial lines extending out of the plot frame show the positions of the nodes of a purely $l=9$ function.

of an $l=7$ function. The wave function in this direction presents an $l=7$ character, as indicated by the nodes (dotted lines) and the radial lines of Fig. 6. Figures 5 and 6 show that the wave function in plane b has an $l=7$ character and is shaped as if it was representing a linear molecule along the x

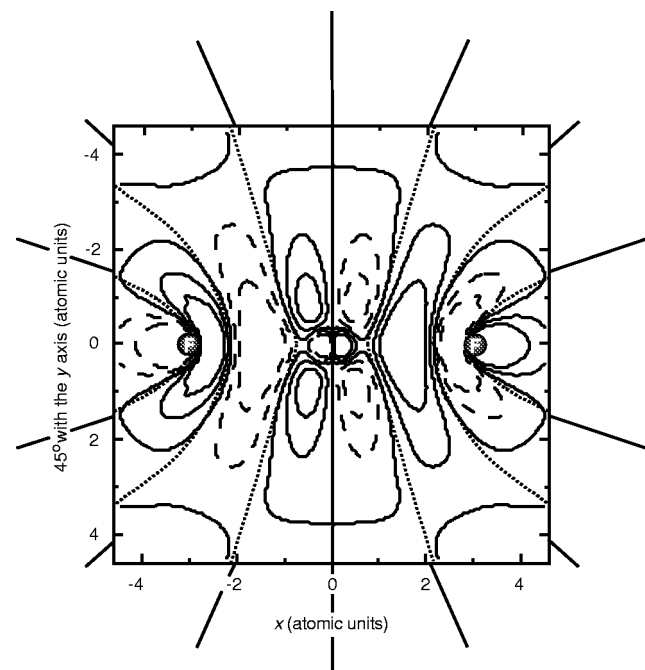


FIG. 6. Contour plot of the ASME resonant state in the plane b of Fig. 7. The lines are labeled as in Fig. 5, except that the radial lines extending out of the plot frame now show the positions of the nodes of a purely $l=7$ function.

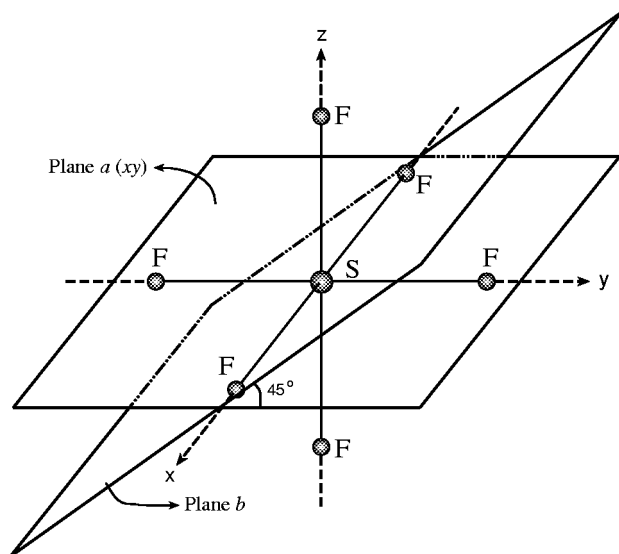


FIG. 7. Planes for the contour plots. Plane a is the xy plane. Plane b contains the x axis and makes 45° with the y direction.

axis, but the presence of the two fluorine atoms along the y axis in plane a introduces two extra nodes so that the complete resonant wave function is a combination of $l=7$ and $l=9$ partial waves. We also note that the contours around the sulfur atom are characteristic of an f wave, which shows the nonvalence nature of this resonance.

In Fig. 8 we see that our asymmetry parameter, as well

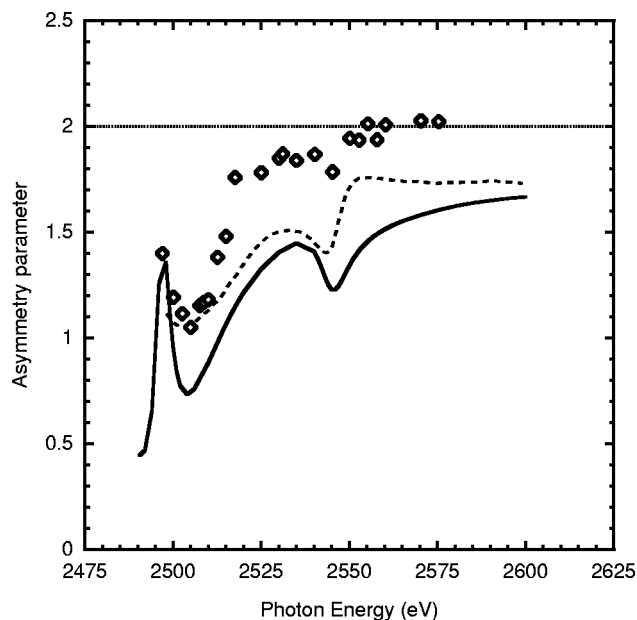


FIG. 8. Asymmetry parameter (β) for $S\,1s$ photoionization of SF_6 . Solid line, present result; dashed line, multiple scattering results (Ref. 24); diamonds, experiment (Ref. 23). The dotted horizontal line shows the maximum theoretical value for β ($\beta=2$).

as the multiple scattering result, shows only a qualitative agreement with the experiment. The quantitative agreement between our curve and the experiment at low energies is only apparent due to the fact that both curves are dropping quickly. Apart from the fact that our result is about 0.3 units too low, our curve reproduces the shape of the experimental data. The experimental minimum at an electron energy of about 2545 eV appears in our result at 2546 eV. The asymmetry parameter is known to be very sensitive to the approximations in the wave function.¹¹ There can be several reasons for the lack of quantitative agreement between our result and the experiment. Some of these reasons are the absence of multichannel coupling, correlation effects, and relaxation of the ionic orbitals in our calculations, and the use of the dipole approximation to describe a process that involves photons with relatively high energy.

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